# Inventory Management

*Viva revision guide — key concepts, formulas, and quick-fire Q&A*

## 1. What is Inventory?

Inventory is the stock of any item or resource used in an organisation. It includes raw materials, work-in-progress (WIP), finished goods, and MRO (maintenance, repair, and operating) supplies. Inventory acts as a buffer between supply and demand, smoothing operations and protecting against uncertainty.

### Types of Inventory

- **Raw materials** — inputs awaiting use in production.
- **Work-in-progress (WIP)** — partially completed goods on the shop floor.
- **Finished goods** — ready-to-ship products.
- **MRO** — consumables for maintaining operations (lubricants, spares).
- **Pipeline / in-transit** — goods moving between stages or locations.

## 2. Why Hold Inventory? (Functions)

- **Decoupling** stages of production so a stoppage in one doesn't halt others.
- **Meeting anticipated demand** (seasonality, promotions).
- **Buffering uncertainty** in demand and supply (safety stock).
- **Economies of scale** via bulk purchasing and longer production runs (cycle stock).
- **Hedging** against price increases or currency fluctuations.
- **Pipeline inventory** for goods in transit.

## 3. Inventory Costs — The Trade-off

> **Core idea:** inventory decisions balance the cost of *ordering / setting up* against the cost of *holding* stock, while keeping stockouts acceptably low. All EOQ-family models are variations on this trade-off.

- **Holding (carrying) cost — H**: storage, insurance, obsolescence, capital tied up. Usually 20–40% of unit cost per year.
- **Ordering / setup cost — S**: paperwork, transport, inspection, machine setup.
- **Purchase / item cost**: unit price (relevant under quantity discounts).
- **Stockout (shortage) cost**: lost sales, backorders, loss of goodwill.

## 4. ABC Analysis (Pareto Classification)

A selective control technique that classifies items by annual rupee usage so that managerial attention is focused where it matters most.

| Class | % of items | % of annual value | Control |
|---|---|---|---|
| A | ~10–20% | ~70–80% | Tight: frequent review, accurate records |
| B | ~30% | ~15–25% | Moderate: periodic review |
| C | ~50–60% | ~5–10% | Simple: bulk orders, two-bin system |

## 5. Economic Order Quantity (EOQ)

EOQ gives the order quantity that minimises total annual inventory cost. Assumes constant demand, constant lead time, no stockouts, and no quantity discounts.

### Key Formulas

```
EOQ = √( 2 × D × S / H )
Number of orders per year = D / EOQ
Time between orders (T) = EOQ / D (in years)
Total annual cost = (D/Q)×S + (Q/2)×H + D×P
```

Where D = annual demand, S = cost per order, H = holding cost per unit per year, P = unit price.

## 6. Reorder Point (ROP) & Safety Stock

$$ROP = d \times L + \text{Safety Stock}$$

$$\text{Safety Stock} = Z \times \sigma_{dLT}$$

d = average daily demand, L = lead time in days, Z = service-level factor (e.g. 1.65 for 95%, 2.33 for 99%), $\sigma_{dLT}$ = std. deviation of demand during lead time. Higher service level ⇒ more safety stock ⇒ higher cost.

## 7. Inventory Control Systems

- **Continuous (Q) system / Fixed-order-quantity**: order a fixed quantity (EOQ) whenever stock falls to ROP. Requires perpetual monitoring.
- **Periodic (P) system / Fixed-time-period**: review stock at fixed intervals and order up to a target level. Simpler but needs higher safety stock.
- **Two-bin system**: a visual reorder trigger — reorder when first bin empties.
- **VED analysis**: Vital, Essential, Desirable — used for spare parts.
- **FSN analysis**: Fast-moving, Slow-moving, Non-moving items.
- **HML analysis**: classification by unit price (High, Medium, Low).

## 8. Modern Approaches

- **Just-In-Time (JIT)**: receive goods only as needed; minimises inventory but demands reliable suppliers and stable demand.
- **Vendor-Managed Inventory (VMI)**: supplier monitors and replenishes buyer's stock.
- **Material Requirements Planning (MRP)**: computes dependent demand from a master production schedule and bill of materials.
- **ERP systems (e.g. SAP)**: integrate inventory with finance, sales, production.
- **RFID & barcoding**: real-time visibility and accuracy.

## 9. Key Performance Indicators

```
Inventory turnover = COGS / Average inventory
Days of supply = 365 / Inventory turnover
Fill rate = Units shipped on time / Units ordered
```

Other KPIs: stockout rate, carrying cost %, shrinkage rate, GMROI.

## 10. Quick-Fire Viva Q&A

**Q. What is the main objective of inventory management?**

A. To make items available when needed at the lowest total cost — balancing customer service against holding, ordering and stockout costs.

**Q. State the assumptions of the basic EOQ model.**

A. Constant demand; constant lead time; entire order received at once; no stockouts; no quantity discounts; only ordering and holding costs vary with quantity.

**Q. Why does EOQ occur where ordering cost equals holding cost?**

A. Because total cost = ordering + holding, and ordering cost decreases while holding cost increases with Q; the sum is minimum where the two curves intersect.

**Q. Difference between independent and dependent demand?**

A. Independent demand comes from the market (finished goods) and is forecast. Dependent demand is derived from independent demand via BOM (e.g. components), and is computed using MRP.

**Q. What is safety stock and what does it depend on?**

A. Extra stock held to protect against variability in demand or lead time. Depends on demand variability, lead-time variability, and the desired service level.

**Q. What is service level?**

A. The probability of not stocking out during the lead time, e.g. 95% means a 5% chance of a stockout per replenishment cycle.

**Q. Why is ABC analysis useful?**

A. It focuses managerial control on the few items that account for most of the value — tight control on A items, lighter control on C items.

**Q. Limitations of EOQ?**

A. Real demand is rarely constant; lead times vary; quantity discounts are common; capacity and shelf-life constraints are ignored.

**Q. What is JIT and what does it require?**

A. Just-In-Time aims for near-zero inventory by receiving inputs just as they are needed. Requires reliable suppliers, short setup times, quality at source, and stable schedules.

**Q. How is inventory turnover interpreted?**

A. Higher turnover means inventory is sold and replenished more frequently — often good, but excessively high turnover can indicate stockout risk.

**Q. What is the bullwhip effect?**

A. Amplification of demand variability as orders move upstream in a supply chain, caused by forecast updating, batching, price fluctuations and rationing.

**Q. Difference between Q-system and P-system?**

A. Q: fixed quantity ordered at variable times when stock hits ROP. P: variable quantity ordered at fixed time intervals, up to a target level.

**Q. How would you reduce inventory in a cafe / small business context?**

A. Improve demand forecasting, shorten supplier lead times, adopt FIFO, classify items (ABC / FSN), reduce SKUs, and use a simple two-bin or par-stock reorder system.

## One-Minute Recap

Inventory exists to **decouple** stages and **buffer uncertainty**. We classify items via **ABC**, decide order size using **EOQ**, decide *when* to order using **ROP = dL + safety stock**, and monitor performance through **turnover, fill rate and days of supply**. Modern systems — JIT, VMI, MRP, ERP — tighten this loop. The eternal trade-off is **holding vs ordering vs stockout** cost.